

Incident Report

Incident Title

SSH Brute Force Attack Detection

Date of Detection

24 February 2026

Severity Level

High

Executive Summary

On 24 February 2026, multiple failed SSH login attempts were detected on server1 from suspicious external IP addresses. The attacker repeatedly attempted to gain unauthorized access to different user accounts including root, admin, and test using brute force techniques. The activity exceeded the allowed authentication attempts, indicating a possible brute force attack targeting the SSH service.

Affected Assets

- Target Server: server1
- Service Targeted: SSH
- Targeted Accounts: root, admin, test
- Suspicious IP Addresses:
 - 185.122.21.7
 - 91.201.45.66

MITRE ATT&CK; Mapping

- T1110 – Brute Force
- T1078 – Valid Accounts

Detailed Timeline of Events

- 10:14:22 – Failed login attempt for invalid user admin from IP 185.122.21.7
- 10:14:27 – Multiple failed root login attempts detected from the same IP
- 10:14:35 – Maximum authentication attempts exceeded for root account
- 10:14:40 – SSH connection closed after repeated failures
- 10:15:02 – Additional failed login attempts detected from IP 91.201.45.66 targeting test account
- 10:15:12 – Connection disconnected due to too many authentication failures

Impact Assessment

- Unauthorized login attempts detected against SSH service
- Multiple user accounts targeted by external attackers
- Potential risk of credential compromise
- Increased risk of unauthorized remote access if successful

Root Cause Analysis

Primary Root Cause:

- Repeated brute force login attempts targeting SSH accounts

Contributing Factors:

- Publicly accessible SSH service
- Weak or guessable usernames
- Lack of IP-based access restrictions

Detection Method

- Splunk log analysis
- Failed authentication monitoring
- Custom brute force detection alert
- SOC dashboard visualization

Actions Performed

- Uploaded SSH logs into Splunk SIEM
- Analyzed failed authentication events
- Identified suspicious attacker IP addresses
- Created brute force detection dashboard
- Configured automated alert for repeated failed logins

Recommended Containment Actions

Immediate Actions

- Block suspicious IP addresses
- Disable unused accounts
- Enforce strong password policies
- Limit SSH login attempts
- Enable multi-factor authentication (MFA)

Long-Term Recommendations

- Implement account lockout policies
- Restrict SSH access using firewall rules
- Enable centralized log monitoring
- Deploy endpoint monitoring solutions
- Conduct regular security audits

Conclusion

The investigation confirmed a brute force attack targeting the SSH service on server1. Multiple failed login attempts were identified from external IP addresses attempting unauthorized access to user accounts. Continuous monitoring, alerting, and access hardening are recommended to prevent future attacks.